Integration by Parts: Choosing the Pieces

Single and repeated integration by parts, and parts with logarithms and inverse trig

- $\displaystyle\int u\,dv = uv - \int v\,du$. Choose u so that differentiating it makes the problem *simpler*, and dv so that integrating it does not make things worse.
- Write your u, du, v and dv before you start. No two problems below use the same skeleton.
- Indefinite answers need $+C$. Give exact values throughout.

1. Evaluate $\displaystyle\int xe^x\,dx$.

Answer: ________________

2. Evaluate $\displaystyle\int \ln x\,dx$. (There is only one factor visible — take $dv = dx$.)

Answer: ________________

3. Evaluate $\int x \sin x \, dx$.

Answer: ________________

4. Evaluate $\int x \ln x \, dx$.

Answer: ________________

5. Evaluate $\int_0^1 x e^{2x} \, dx$. Give an exact value.

Answer: ________________

6. Consider $\int x^2 e^x \, dx$.

(a) Evaluate it.

Answer: ____________________

(b) Explain why parts has to be applied twice here.

Answer: ____________________

7. Evaluate $\int \arctan x \, dx$. (Again there is only one factor — take $dv = dx$.)

Answer: ____________________

8. Evaluate $\int e^x \sin x \, dx$. Apply parts twice and solve for the integral that comes back.

Answer: __________________

9. Evaluate $\int_0^{\pi} x \cos x \, dx$.

Answer: __________________

10. Consider $\int_0^b xe^{-x}\,dx$.

(a) Evaluate it for $b = 3$. Give an exact value.

Answer: ____________________

(b) In general the integral equals $1 - (b+1)e^{-b}$. Find its limit as $b \to \infty$.

Answer: ____________________

(c) Explain why taking $u = x$ rather than $u = e^{-x}$ is the strategic choice here.

Answer: ____________________